

The Network Origins of Inflation Stances in a Currency Union

Zhihao Xu

Emory University

August 19, 2026

Research Questions

- **Questions**

- What structural forces shape cross-country divergence in inflation preferences in a currency union?
- What's the welfare implication of this divergence?

- **This paper:** proposes a production-network explanation.

- Studies optimal policy in a heterogeneous-agent input-output (HAIO) economy.
- Defines *unilateral inflation stance* and *policy-alignment loss* (PAL).
- Quantitatively: a country's inflation stance operates mainly through its *input-output multiplier*, and hence, relates to its position in the union's network.

Related Literature

- New Open Economy Macroeconomics (NOEM) literature on **currency-union policy design**: e.g., Benigno, 2004; Galí, 2008; Ferrero, 2009; Farhi and Werning, 2017.
- HANK literature on the **redistribution motive of monetary policy**: e.g., Bhandari et al., 2021; Acharya et al., 2023; Dávila and Schaab, 2023; Le Grand et al., 2022; La'O and Morrison, 2024; Nuño and Thomas, 2022.
- Production network literature on **optimal monetary policy**: e.g. La'O and Tahbaz-Salehi, 2022; Rubbo, 2023; Xu and Yu, 2025; Qiu et al., 2026.

Environment

- The framework of this paper sits at the intersection of Baqaee and Farhi (2024) and La'O and Tahbaz-Salehi (2022).
- Static New Keynesian economy with C countries and N industries.
- Each country has a representative household that supplies a **distinct type of labor**.
- Countries differ in labor supply elasticities, sectoral labor input shares, ownership structures, and consumption baskets.

Firms

- A unit mass of monopolistically competitive firms $k \in [0, 1]$ in sector i

$$y_{ik} = z_i F_i(\{x_{ij,k}\}_{j=1}^N, \{L_{ic}\}_{c=1}^C) = z_i \varsigma_i \prod_{j=1}^N x_{ij,k}^{\omega_{ij}} \prod_{c=1}^C L_{ic,k}^{\alpha_{ic}}$$

- In the calibration (geographic segmentation), α_{ic} equals the value-added share of industry i if it is located in country c and zero otherwise.
- Differentiated varieties within an industry i aggregated via a CES aggregator

$$y_i = \left(\int_0^1 y_{ik}^{\frac{\theta_i-1}{\theta_i}} dk \right)^{\frac{\theta_i}{\theta_i-1}}$$

- All firms in industry i minimize

$$mc_i = \min_{\{x_{ij,k}\}, \{L_{ic,k}\}} \sum_{c=1}^C w_c L_{ic,k} + \sum_{j=1}^N p_j x_{ij,k}, \quad s.t. \ y_{ik} = 1$$

Nominal Rigidities

- Firm k in industry i can reset price with probability $\delta_i \in (0, 1]$. The optimal reset price maximizes the profit

$$\Pi_{ik} = \max_{p_{ik}} [p_{ik} - (1 - \tau_i)mc_i] \left(\frac{p_{ik}}{p_i} \right)^{-\theta_i} y_i$$

- To a first-order approximation,

$$\log \mathbf{p} = \delta \log \mathbf{mc}, \quad \log \boldsymbol{\mu} = -(\delta^{-1} - I) \log \mathbf{p}$$

where $\delta = \text{diag}(\delta_1, \dots, \delta_N)$.

Households

- Utility function of household c

$$U_c(C_c, L_c) = \log(C_c) - \psi_c \frac{L_c^{1+1/\eta_c}}{1 + 1/\eta_c}$$

where C_c is a utility-relevant aggregator of sectoral goods consumed by household c

$$C_c = \prod_{i=1}^N (c_{ci}/\beta_{ci})^{\beta_{ci}}.$$

- Budget constraint

$$\text{GNE}_c \equiv \sum_{i=1}^N p_i c_{ci} \leq w_c L_c + \sum_{i=1}^N \Phi_{ic} \Pi_i - T_c$$

where Φ is the ownership matrix (geographic segmentation: $\Phi_{ic} = \mathbf{1}\{i \in N_c\}$).

Policy Instruments

- **Fiscal policy:** industry subsidies $1 - \tau_i = (\theta_i - 1)/\theta_i$ eliminate CES distortions; financed by lump-sum taxes proportional to ownership shares,

$$T_c = \sum_{i \in N} \Phi_{ic} \tau_i m c_i \int_0^1 y_{ik} dk.$$

- **Monetary policy:** controls nominal aggregate demand

$$\text{GNE} = \sum_{c=1}^C \sum_{i=1}^N p_i c_{ci} = m.$$

Equilibrium Definition

Definition

A *sticky-price equilibrium* is a set of allocations, prices, and policies such that:

- (i) firms optimally choose inputs and reset prices when allowed;
- (ii) each household maximizes utility subject to its budget constraint;
- (iii) the government budget constraint is satisfied;
- (iv) all markets clear.

Definition

The *flexible-price equilibrium* is defined by the same conditions, except that all firms reset their prices optimally.

- Log deviation from the flexible-price equilibrium: $\hat{x} = \log x - \log x^*$.

Minimum Notations

	Country c	Union
Income share	$\chi_c = \frac{\text{GNE}_c}{\text{GNE}}$	$\sum_c \chi_c = 1$
Consumption share	β_{ci}	$b_i = \sum_c \chi_c \beta_{ci}$
Domar weight	$\lambda_i^c = \sum_j \beta_{cj} \Psi_{ji}$	$\lambda_i = \sum_c \chi_c \lambda_i^c$

- $\Psi = (I - \Omega)^{-1}$ is the Leontief inverse, where $\Omega = [\omega_{ij}]$ is the input–output matrix.
- β_{ci} is **direct** exposure of country c to industry i .
- λ_i^c is **direct** and **indirect** exposure of country c to industry i .

Allocative Efficiency

Proposition

To a first-order approximation,

$$\hat{c}_c - \hat{l}_c = \underbrace{\mathcal{J}_c^{\text{DI}'} \log \mu}_{\text{direct incidence}} + \underbrace{\mathcal{J}_c^{\text{FToT}'} \log \mu}_{\text{factoral terms of trade}} \equiv \mathcal{J}_c' \log \mu.$$

- **Direct incidence** reflects a country's dual role as both a consumer and a profit owner (supplier):

$$\Delta \text{DI}_c = \mathcal{J}_c^{\text{DI}'} \log \mu = \sum_i \left(\frac{\lambda_i \Phi_{ic}}{\chi_c} - \lambda_i^c \right) \log \mu_i$$

- **Factor terms of trade** measures the relative value of the labor a country supplies to the labor costs it faces:

$$\Delta \text{FToT}_c = \mathcal{J}_c^{\text{FToT}'} \log \mu = d \log w_c - \sum_f \Lambda_f^c d \log w_f$$

Allocative Efficiency as Overall Terms of Trade

- In a horizontal economy (without input–output linkages), and assuming ownership segmentation,

$$\Delta DI_c = \mathcal{J}_c^{DI'} \log \mu = \underbrace{\sum_{i \in N_c} \sum_{f \neq c} \chi_f \beta_{fi} / \chi_c \log \mu_i}_{\text{change in export prices}} - \underbrace{\sum_{i \notin N_c} \beta_{ci} \log \mu_i}_{\text{change in import prices}} .$$

- The direct incidence effect** maps to the commodity terms of trade effect.
- Allocative efficiency** = overall (commodity + factoral) terms of trade effect.

The Role of Production Networks

Input–Output Multiplier

$$\xi_c \equiv \sum_{i \in N_c} \frac{\lambda_i}{\chi_c} = \frac{\sum_{i \in N_c} p_i y_i}{\text{GNE}_c}$$

measures total domestic sales relative to domestic final demand.

- Under ownership segmentation,

$$\Delta \text{DI}_c = \underbrace{(\xi_c - \mathbb{E}_\chi[\xi]) \overline{\log \mu}}_{\text{IO-multiplier channel}} + \underbrace{\left(\mathbb{E}_\chi[\xi] - \sum_i \lambda_i^c \right) \overline{\log \mu}}_{\text{home-bias channel}} + \underbrace{\sum_i \mathcal{J}_{ic}^{\text{DI}} (\log \mu_i - \overline{\log \mu})}_{\text{sectoral-heterogeneity channel}}.$$

- Without production networks,

$$\xi_c = 1, \quad \sum_i \lambda_i^c = 1,$$

so the first two channels disappear.

- When the IO-multiplier channel dominates, countries with above-average ξ_c experience a deterioration in their terms of trade following a monetary expansion ($\overline{\log \mu} < 0$).

Aggregate Cancellation: A Macro Envelope Theorem

Corollary

Income-share-weighted allocative efficiencies exactly cancel at first order:

$$\sum_{c=1}^C \chi_c (\hat{c}_c - \hat{l}_c) = \sum_{c=1}^C \chi_c \mathcal{J}'_c \log \mu = 0.$$

- **Intuition:** a closed economy has zero aggregate terms of trade.
- **Implication:** monetary policy has a redistribution motive whenever welfare aggregation departs from income-share weighting.

Bergson–Samuelson Welfare

- Adopt the Bergson–Samuelson welfare aggregator

$$W(\{\kappa_c\}_{c=1}^C) = \sum_{c=1}^C \kappa_c U_c,$$

where $\{\kappa_c\}$ are Pareto weights satisfying $\sum_c \kappa_c = 1$.

Proposition (Bergson–Samuelson Welfare function)

For arbitrary Pareto weights $\{\kappa_c\}_{c=1}^C$,

$$W(\{\kappa_c\}) - W^*(\{\kappa_c\}) = \sum_{c=1}^C \kappa_c \mathcal{J}'_c \log \mu - \frac{1}{2} \log \mu' \mathcal{L}(\{\kappa_c\}) \log \mu.$$

- Generalizes Baqaee and Farhi (2024) (income-share weighting) and La'O and Tahbaz-Salehi (2022) (representative agent) to a HAI0 economy with arbitrary Pareto weights.
- Under income-share weighting, monetary policy has a purely efficiency objective, $\sum_c \chi_c \mathcal{J}'_c \log \mu = 0$.
- Takeaway:** monetary policy faces a tradeoff between **aggregate stabilization** and **first-order redistribution** whenever Pareto weights deviate from income shares.

Optimal Policy

Proposition (Optimal Monetary Policy (Second Best))

Given Pareto weights $\{\kappa_c\}_{c=1}^C$, the optimal monetary policy targets the sectoral price index

$$\sum_{j=1}^N \zeta_j^*(\{\kappa_c\}) \log p_j = \pi^*(\{\kappa_c\})$$

with sectoral weights and optimal inflation bias

$$\zeta_j^*(\{\kappa_c\}) = (\delta_j^{-1} - 1) \sum_{i=1}^N (\delta_i^{-1} - 1) \varrho_i^m \mathcal{L}(\{\kappa_c\})_{ij},$$

$$\pi^*(\{\kappa_c\}) = \sum_{c=1}^C \kappa_c \mathcal{J}'_c(I - \delta^{-1}) \varrho^m.$$

Here, $\varrho^m \equiv \frac{d \log p}{d \log m} = \left[I - \varrho^w (\Gamma + \ell^\mu \eta^{-1})' (I - \delta^{-1}) \right]^{-1} \varrho^w \mathbf{1}$ denotes sectoral monetary pass-through.

Centralized vs. Unilateral Optimal Policy

	Centralized	Unilateral
Pareto weights	$\kappa = \chi$	$\kappa = \mathbf{e}_c$ ($\kappa_c = 1$)
Objective	Minimize union-wide welfare loss	Minimize country c 's welfare loss
Optimal policy	$\sum_j \zeta_j^*(\chi) \log p_j = 0$	$\sum_j \zeta_j^*(\mathbf{e}_c) \log p_j = \pi^*(\mathbf{e}_c)$
Inflation target	Zero	$\pi^*(\mathbf{e}_c)$

- $\pi^*(\mathbf{e}_c)$ measures how country c 's **terms of trade** respond to a monetary expansion.
- Country c 's unilateral optimal policy is inflationary if $\pi^*(\mathbf{e}_c) > 0$, indicating that its terms of trade improve following a monetary expansion.

Unilateral Inflation Stance

- The unilateral optimal policy

$$(\zeta^*(\mathbf{e}_c), \pi^*(\mathbf{e}_c)) = \arg \min_{(\zeta, \pi)} \mathbb{L}_c(\zeta, \pi)$$

assumes country c fully controls monetary policy.

- In a currency union, however, the target price index ζ is determined collectively (e.g., by the ECB Governing Council). Each country can only express its preferred inflation rate.

Definition

Given the union's target price index ζ , country c 's **inflation stance** is

$$\pi_c(\zeta) = \arg \min_{\pi \in \mathbb{R}} \mathbb{E}[\mathbb{L}_c(\zeta, \pi)].$$

Proposition

The unilateral inflation stance satisfies

$$\pi_c(\zeta) = \underbrace{\frac{\zeta' \varrho^m}{[\zeta^*(\mathbf{e}_c)]' \varrho^m}}_{\text{price-index adjustment}} \pi^*(\mathbf{e}_c).$$

Interpretation. $\pi_c(\zeta)$ is a rescaled version of $\pi^*(\mathbf{e}_c)$.

Policy-Alignment Loss (PAL)

Definition (Policy-Alignment Loss)

The *policy-alignment loss* for country c under a union-wide policy (ζ, π) is

$$\text{PAL}_c(\zeta, \pi) \equiv \mathbb{L}_c(\zeta, \pi) - \mathbb{L}_c(\zeta^*(\mathbf{e}_c), \pi^*(\mathbf{e}_c)) \geq 0.$$

Proposition (PAL Decomposition)

The expected PAL admits the closed-form decomposition

$$\mathbb{E}[\text{PAL}_c(\zeta, \pi)] = \underbrace{\frac{1}{2} \frac{\zeta^*(\mathbf{e}_c)' \varrho^m}{(\zeta' \varrho^m)^2} [\pi_c(\zeta) - \pi]^2}_{\text{inflation misalignment}} + \underbrace{\mathcal{I}_c(\zeta)}_{\text{price-index misalignment}},$$

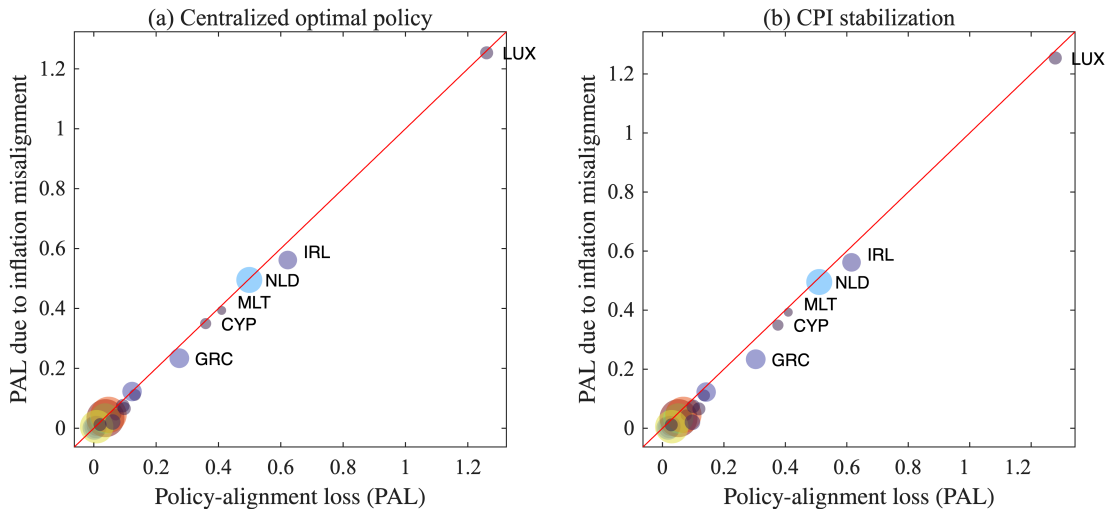
where $\mathcal{I}_c(\zeta) \geq 0$ vanishes at $\zeta = \zeta^*(\mathbf{e}_c)$.

- Inflation misalignment is **quadratic** in the distance $|\pi_c(\zeta) - \pi|$.

Calibration

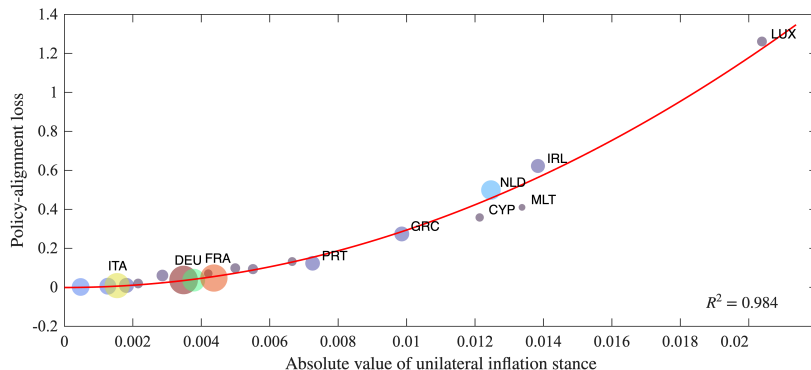
- **Parameters:** $\eta_c = 2$, $\theta_i = 6$, $\Phi_{ic} = \mathbf{1}\{i \in N_c\}$ (geographically segmented ownership).
- **IO matrix:** WIOD 2014 (Timmer et al., 2015); $C = 20$ euro-area countries, each with $N_c = 56$ industries.
- **Price rigidities:** monthly FPA from Pastén et al. (2020), mapped to WIOD industries via NAICS; converted to quarterly.
- **Sectoral productivity shocks:** annual TFP growth from WIOD SEA via Törnqvist index, interpolated to quarterly, linearly detrended, and used to form the variance–covariance matrix.

Decomposition of Policy-Alignment Loss



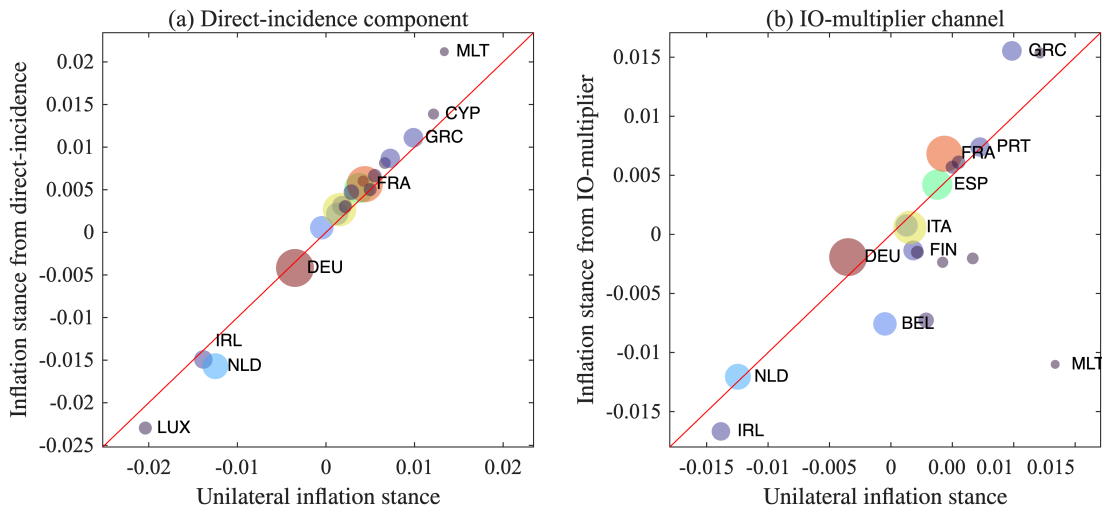
- PAL is mainly driven by inflation misalignment (quadratic in inflation stance deviation).

Inflation-Stance Deviation and PAL: Quadratic Fit



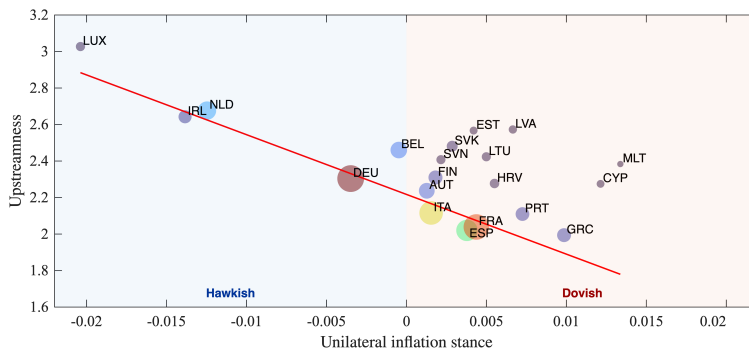
- Union-wide inflation consensus is zero under the centralized optimal policy.
- Quadratic fit (no linear term) gives $R^2 = 0.98$.
- A one-percentage-point inflation deviation predicts a loss of about 0.29 pp of steady-state consumption.
- Tail countries (LUX, IRL, NLD, GRC) bear disproportionate welfare costs.

Decomposition of the Inflation Stance



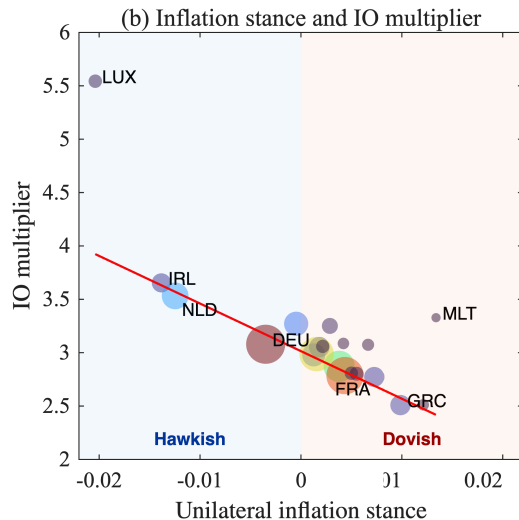
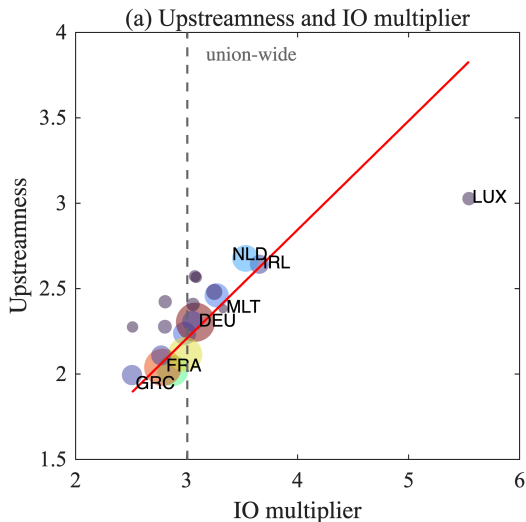
- Panel (a): The direct-incidence component reproduces almost the entire unilateral inflation stance.
- Panel (b): the *IO-multiplier channel* alone reproduces most cross-country variation.

Inflation Stance and Upstreamness



- **Upstreamness** (Antràs et al., 2012): value-added-weighted distance from final consumption.
- Strong negative relationship: WLS slope ≈ -32.7 , $R^2 = 0.77$.
- More upstream countries prefer lower inflation; downstream prefer higher.
- Mirrors the familiar North–South divide in the euro area.

IO Multiplier as Mediating Network Statistic



- Panel (a): upstreamness $\leftrightarrow$ local IO multiplier ($R^2 = 0.77$).
- Panel (b): IO multiplier $\leftrightarrow$ unilateral inflation stance ($R^2 = 0.75$).
- Mechanism: network position $\Rightarrow$ IO multiplier $\Rightarrow$ terms of trade $\Rightarrow$ inflation preference.

Conclusion

- This paper develops a HAIO model and provides microfoundations for divergent inflation preferences across member states in a currency union.
- Bringing the model to the euro area, I find that a country's inflation stance negatively relates to its upstreamness, and the deviation from the union-wide consensus predicts welfare costs.
- Therefore, countries at either extreme of the production network, whether highly upstream or highly downstream, incur the largest welfare costs.

References

- Acharya, Sushant, Edouard Challe, and Keshav Dogra (2023) 'Optimal monetary policy according to HANK.' *American Economic Review* 113(7), 1741–1782
- Antràs, Pol, Davin Chor, Thibault Fally, and Russell Hillberry (2012) 'Measuring the upstreamness of production and trade flows.' *American Economic Review* 102(3), 412–416
- Baqaei, David R., and Emmanuel Farhi (2024) 'Networks, Barriers, and Trade.' *Econometrica* 92(2), 505–541
- Benigno, Pierpaolo (2004) 'Optimal monetary policy in a currency area.' *Journal of International Economics* 63(2), 293–320
- Bhandari, Anmol, David Evans, Mikhail Golosov, and Thomas J Sargent (2021) 'Inequality, Business Cycles, and Monetary-Fiscal Policy.' *Econometrica* 89(6), 2559–2599
- Dávila, Eduardo, and Andreas Schaab (2023) 'Optimal monetary policy with heterogeneous agents: Discretion, commitment, and timeless policy.' Technical Report, National Bureau of Economic Research
- Farhi, Emmanuel, and Iván Werning (2017) 'Fiscal unions.' *American Economic Review* 107(12), 3788–3834
- Ferrero, Andrea (2009) 'Fiscal and monetary rules for a currency union.' *Journal of International Economics* 77(1), 1–10
- Galí, Jordi (2008) *Monetary policy, inflation, and the business cycle: an introduction to the new Keynesian framework* (Princeton University Press)
- La'O, Jennifer, and Alireza Tahbaz-Salehi (2022) 'Optimal monetary policy in production networks.' *Econometrica* 90(3), 1295–1336
- La'O, Jennifer, and Wendy A Morrison (2024) 'Optimal monetary policy with redistribution.' Technical Report, National Bureau of Economic Research
- Le Grand, François, Alaïs Martin-Baillon, and Xavier Ragot (2022) 'Should monetary policy care about redistribution? Optimal monetary and fiscal policy with heterogeneous agents'
- Nuño, Galo, and Carlos Thomas (2022) 'Optimal redistributive inflation.' *Annals of Economics and Statistics* (146), 3–64
- Pastén, Ernesto, Raphael Schoenle, and Michael Weber (2020) 'The propagation of monetary policy shocks in a heterogeneous production economy.' *Journal of Monetary Economics* 116(C), 1–22
- Qiu, Zhesheng, Yicheng Wang, Le Xu, and Francesco Zanetti (2026) 'Monetary policy in open economies with production networks.' *Journal of Monetary Economics* 159, 103918
- Rubbo, Elisa (2023) 'Networks, Phillips curves, and monetary policy.' *Econometrica* 91(4), 1417–1455
- Timmer, Marcel P., Erik Dietzenbacher, Bart Los, Robert Stehrer, and Gaaitzen J. Vries (2015) 'An illustrated user guide to the world input–output database: The case of global automotive production.' *Review of International Economics* 23(3), 575–605
- Xu, Zhihao, and Changhua Yu (2025) 'Optimal monetary policy in production networks with distortions.' *Journal of Economic Theory* 225, 105979